Biochemistry Cumulative Exam
Protein Synthesis Total Points: 100

1. What biomolecule is the catalytic part of the ribosome? (5 points)

2. What is the energy source required to add amino acids to tRNA? And what class of enzymes perform this function (10 points)

3. How do ribosomes ensure that the correct tRNA is being selected for by the ribosome? (10 points)

4. Protein synthesis or translation requires 4 phases to complete. Describe how each phase contributes to the translational cycle. (10 points)

5. Draw the chemical reaction of peptide bond formation via electron pushing, using gly-$tRNA^{gly}$ and ala-$tRNA^{ala}$ as substrates. (10 points) (supplemental figure 1).

6. Translation requires energy in the form of GTP (supplemental figure 2). How many GTP would be required to generate a 10 amino acid peptide? Describe where each GTP is being hydrolyzed (15 points)

7. The mRNA for SARS-CoV-2 can be translated in the 0 frame or a -1 frame (supplemental figure 3). Why does the virus use this mechanism to encode proteins and what features of the SARS-CoV-2 mRNA allow for it to occur? (15 points)

8. Roughly ½ of all antibiotics target ribosomes. Why are ribosomes good antibiotic targets. (10 points)

9. If you were to design an mRNA from scratch that could be translated by eukaryotic ribosomes for a potential vaccine, what would you need to include and why? (15 points)

Figure 1

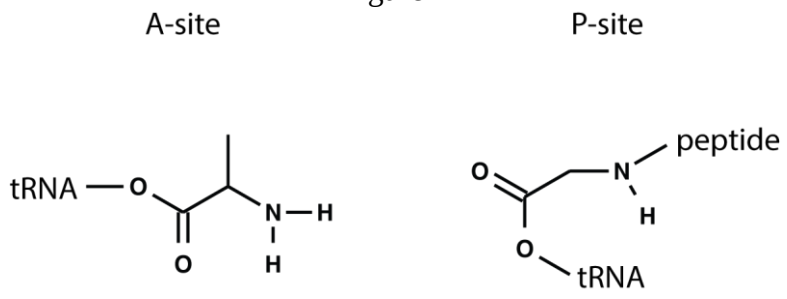
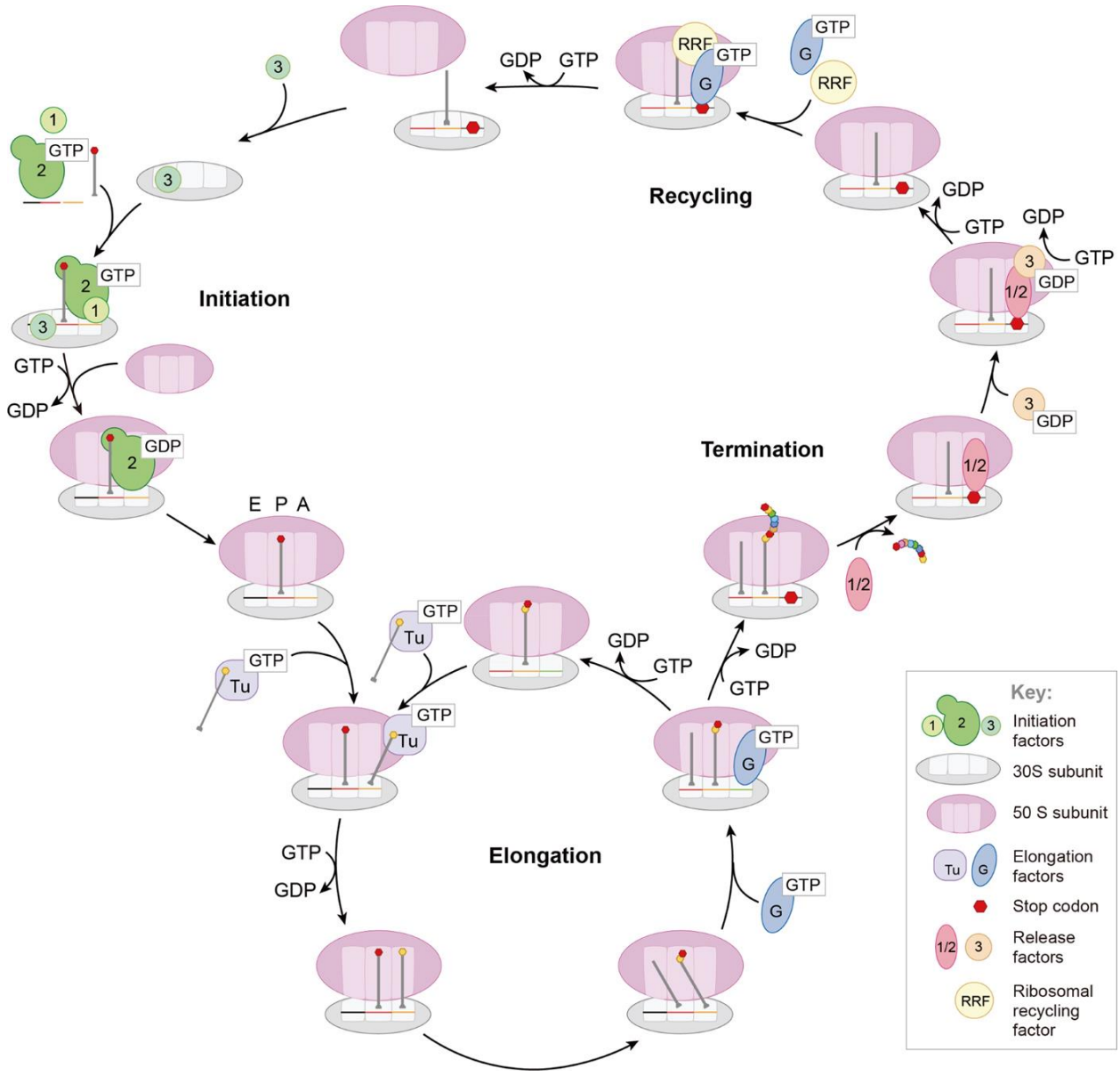
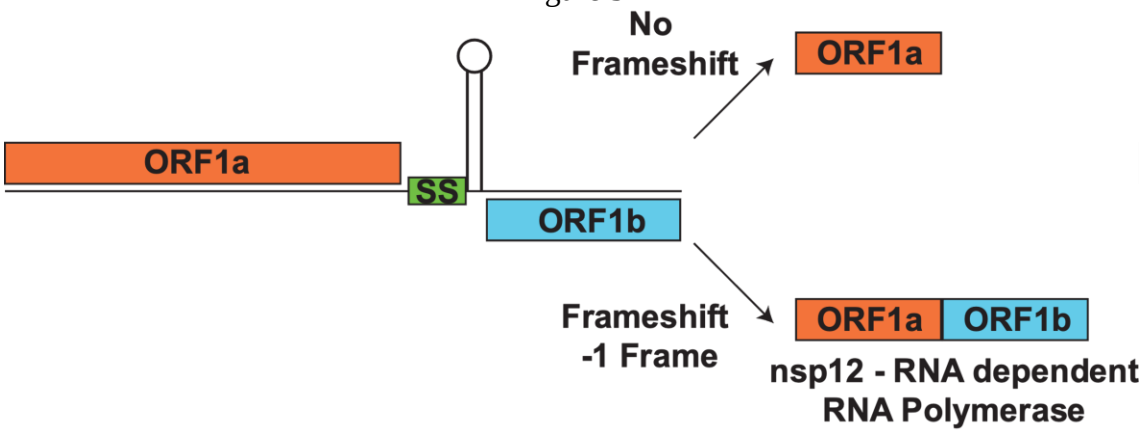


Figure 2



Chen *et al.*, Quarterly Reviews in Biophysics, 2016.

Figure 3



ORF – Open reading frame
SS – slippery sequence
Nsp12 – SARS-CoV-2 RNA dependent RNA polymerase